

ISSUE / BUG LOG

AlgoNova – AI-Powered DSA & System Design Mastering Platform

Verified issues extracted from the AlgoNova technical audit. Historical defects are separated from known limitations and potential security risks.

1. Verified Historical Issues

Issue ID	Module	Description	Type	Severity	Root Cause	Fix	Status	Date
BUG-001	CI/CD / Deployment	Vercel deployment blocked on push.	Deployment	High	Vercel Hobby Plan deployment restrictions caused a mismatch between commit identity and authenticated owner credentials.	Reverted affected commit and pushed a clean commit using authenticated owner credentials.	Resolved	30 Sep 2026
BUG-002	Backend Testing	ECONNREFUSED occurred during backend test-suite execution.	Functional / Testing	Medium	Jest/Supertest attempted localhost:5000 while the Express server was not running during the test phase.	Reconfigured tests to mock the network layer or pass the Express app directly to Supertest.	Resolved	30 Sep 2026
BUG-003	Practice HQ / Frontend	Code editor disappeared after submission.	UI/UX	High	Voice AI container expanded without a fixed height, pushing Monaco Editor outside the visible viewport.	Reworked layout into a flexbox split pane and removed the legacy voice UI.	Resolved	07 Oct 2026
BUG-004	Profile / Frontend	Activity heatmap failed to render correctly.	Dependency	Low	react-activity-calendar was deprecated/incompatible with React 19.	Removed the dependency and implemented a custom MonthSeparatedHeatmap component.	Resolved	15 Jun 2026

2. Known Limitations and Potential Risks

ID	Module	Description	Type	Severity	Root Cause / Technical Basis	Impact	Status
RISK-001	Backend / Code Execution	User-submitted code is compiled and executed directly on the host through child_process.exec without a secure sandbox.	Security Risk	Critical	executor.js uses host-level process execution rather than Docker/serverless isolation.	Malicious submissions could potentially execute shell commands and lead to Remote Code Execution.	Open / Architectural mitigation required
LIM-001	Voice Interview	Voice features may fail on Firefox and Safari.	Compatibility Limitation	Medium	Implementation relies on the non-standard window.webkitSpeechRecognition API.	Users on unsupported browsers may be unable to use browser-based voice interview functionality.	Known Limitation

3. Issue Statistics

Category	Count
Verified historical bugs	4
Resolved verified bugs	4
Open verified historical bugs	0
Critical potential security risks	1
Known compatibility limitations	1

4. Corrective Action Summary

The documented defects were addressed through deployment configuration correction, test-environment reconfiguration, frontend layout restructuring, and replacement of an incompatible third-party dependency. The remaining unrestricted local code-execution risk is architectural and should be addressed through secure isolation before production use.

5. Reporting Note

The log distinguishes verified historical bugs from potential risks and known limitations. No potential defect has been presented as a historical bug without supporting evidence.